PHYS 143 – Mechanics and Thermal Physics

# Week 2: Motion – Tutorial Sheet

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Use IDENTIFY, SET UP, EXECUTE, EVALUATE. Sketch and predict before you calculate; neglect air resistance and use $g = 9.80\,\mathrm{m/s^2}$. Problems A1 and A2 were moved out of the lecture; full solutions follow them. Final answers to the others are at the end.

## A. Moved from the lecture

### A1. A train slows down

A train slows from $30.0\,\mathrm{m/s}$ to $10.0\,\mathrm{m/s}$ while travelling $200\,\mathrm{m}$ along a straight track. Find the (constant) acceleration and the time taken.

**Solution.** Take $+x$ along the motion. The time is missing, so use $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$:

$$a_x = \frac{10.0^2 - 30.0^2}{2(200)} = -2.00\,\mathrm{m/s^2}.$$

Then $t = (v_x - v_{0x})/a_x = (10.0 - 30.0)/(-2.00) = 10.0\,\mathrm{s}$. *Check:* the average speed is $20.0\,\mathrm{m/s}$, and $20.0 \times 10.0 = 200\,\mathrm{m}$. The sign $a_x < 0$ agrees with slowing down.

### A2. Police car catching a motorist

A motorist travelling at a constant $15.0\,\mathrm{m/s}$ passes a stationary police car. At that instant ($t = 0$) the police car starts with constant acceleration $3.0\,\mathrm{m/s^2}$. When and where does it catch the motorist, and how fast is it then moving?

**Solution.** Origin at the police car, $+x$ along the road. Same position at the same time: $15.0\,t = \frac{1}{2}(3.0)t^2$, so $t(1.5t - 15.0) = 0$, giving $t = 0$ or $t = 10\,\mathrm{s}$. The root $t = 0$ is the instant the motorist passes; the physical answer is $t = 10\,\mathrm{s}$. The position is $15.0 \times 10 = 150\,\mathrm{m}$ and the police speed is $3.0 \times 10 = 30\,\mathrm{m/s}$, twice the motorist's. *Check:* both $x_M$ and $x_P$ equal $150\,\mathrm{m}$ at $10\,\mathrm{s}$.

## B. Practice

### B1. Graphs and calculus

A particle on the $x$-axis has $v_x = 12 - 4t$ (SI units), with $x = 0$ at $t = 0$. (a) Find $a_x$. (b) When is it momentarily at rest? (c) Find its displacement and the distance travelled from $0$ to $5.0\,\mathrm{s}$. (d) Sketch $x$–$t$, $v_x$–$t$ and $a_x$–$t$.

### B2. Free fall

(a) A ball is dropped from a $45.0\,\mathrm{m}$ tower. When does it land and how fast is it moving? (b) A second ball is thrown upward from the ground at $20.0\,\mathrm{m/s}$. Find its maximum height, the time to reach it, and the time to return.

### B3. Projectile from a cliff

A stone is thrown horizontally at $15.0\,\mathrm{m/s}$ from the top of a $45.0\,\mathrm{m}$ cliff. Find the time of flight, the horizontal distance travelled, and the speed and direction of the stone when it lands.

### B4. Circular motion

The Moon orbits the Earth in a nearly circular path of radius $3.84 \times 10^{8}\,\mathrm{m}$ with period $27.3$ days. Find its speed and the magnitude of its acceleration.

### B5. Relative velocity

An aircraft has an airspeed of $200\,\mathrm{km/h}$. A wind blows toward the east at $60\,\mathrm{km/h}$. (a) If the pilot points the aircraft due north, find its velocity relative to the ground. (b) In what direction must the pilot point to fly due north over the ground, and what is the ground speed then?

### B6. Challenge: the method is not given

A passenger in a train moving at $20.0\,\mathrm{m/s}$ east throws a ball straight up, at $8.0\,\mathrm{m/s}$ relative to the train. Describe the motion as seen from the train and from the ground. Where does the ball land relative to the passenger, and how far along the track has it travelled relative to the ground?

## Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80\,\mathrm{m/s^2}$.

**E1.** The position of a particle is $x = 2.5t^3 - 4.0t^2 + 3.0t$ ($x$ in m, $t$ in s). Its acceleration at $t = 1.2\,\mathrm{s}$ is

**(A)** $4.2\,\mathrm{m/s^2}$

**(B)** $8.0\,\mathrm{m/s^2}$

**(C)** $9.0\,\mathrm{m/s^2}$

**(D)** $10.0\,\mathrm{m/s^2}$

**(E)** $18.0\,\mathrm{m/s^2}$

**Items E2 and E3** share this information: A car increases its speed uniformly from $8.0\,\mathrm{m/s}$ to $20.0\,\mathrm{m/s}$ while travelling $150\,\mathrm{m}$ in a straight line.

**E2.** The magnitude of its acceleration is

**(A)** $0.56\,\mathrm{m/s^2}$

**(B)** $1.12\,\mathrm{m/s^2}$

**(C)** $1.33\,\mathrm{m/s^2}$

**(D)** $1.60\,\mathrm{m/s^2}$

**(E)** $2.24\,\mathrm{m/s^2}$

**E3.** The time taken for this change of speed is

**(A)** $7.5\,\mathrm{s}$ **(B)** $10.7\,\mathrm{s}$ **(C)** $12.0\,\mathrm{s}$ **(D)** $18.8\,\mathrm{s}$ **(E)** $21.4\,\mathrm{s}$

**E4.** A stone is thrown horizontally at $12.0\,\mathrm{m/s}$ from the top of a $45.0\,\mathrm{m}$ cliff. Ignoring air resistance, it lands at what horizontal distance from the foot of the cliff?

**(A)** $18.2\,\mathrm{m}$ **(B)** $25.7\,\mathrm{m}$ **(C)** $36.4\,\mathrm{m}$ **(D)** $55.1\,\mathrm{m}$ **(E)** $110.2\,\mathrm{m}$

**E5.** A ball is kicked at $18.0\,\mathrm{m/s}$ at $35^\circ$ above the horizontal over level ground. Ignoring air resistance, its maximum height is

**(A)** $5.44\,\mathrm{m}$ **(B)** $7.30\,\mathrm{m}$ **(C)** $9.48\,\mathrm{m}$ **(D)** $11.09\,\mathrm{m}$ **(E)** $16.53\,\mathrm{m}$

**E6.** Which of the following is/are true for a particle in uniform circular motion of radius $r$ and speed $v$?

I. Its velocity is constant because its speed is constant.
II. Its acceleration has magnitude $v^2/r$ and points toward the centre.
III. Its period is $T = 2\pi r/v$.

**(A)** I only **(B)** I and II only **(C)** I and III only **(D)** II and III only **(E)** I, II and III

**E7.** A boat heads straight across a $120\,\mathrm{m}$ wide river at $4.0\,\mathrm{m/s}$ relative to the water. The current is $2.0\,\mathrm{m/s}$ downstream. How far downstream does the boat land?

**(A)** $15\,\mathrm{m}$ **(B)** $30\,\mathrm{m}$ **(C)** $54\,\mathrm{m}$ **(D)** $60\,\mathrm{m}$ **(E)** $120\,\mathrm{m}$

**E8.** A particle starts from rest with acceleration $\vec{a} = (6.0t\,\hat{\boldsymbol{\imath}} - 2.0\,\hat{\boldsymbol{\jmath}})\,\mathrm{m/s^2}$. Its speed at $t = 2.0\,\mathrm{s}$ is

**(A)** $12.6\,\mathrm{m/s}$

**(B)** $10.0\,\mathrm{m/s}$

**(C)** $12.0\,\mathrm{m/s}$

**(D)** $12.2\,\mathrm{m/s}$

**(E)** $16.0\,\mathrm{m/s}$

**E9.** A particle has $x = 3.0t^2 - 12.0t + 5.0$ ($x$ in m, $t$ in s). Its position when it is momentarily at rest is

**(A)** $-19\,\mathrm{m}$ **(B)** $-12\,\mathrm{m}$ **(C)** $-10\,\mathrm{m}$ **(D)** $-9\,\mathrm{m}$ **(E)** $-7\,\mathrm{m}$

**E10.** A ball is thrown vertically upward at $14.0\,\mathrm{m/s}$ from the ground. It returns to the ground after

**(A)** $1.43\,\mathrm{s}$ **(B)** $2.86\,\mathrm{s}$ **(C)** $4.29\,\mathrm{s}$ **(D)** $5.71\,\mathrm{s}$ **(E)** $10.0\,\mathrm{s}$

## Answers

B1 (a) $-4.0\,\mathrm{m/s^2}$; (b) $t = 3.0\,\mathrm{s}$; (c) displacement $+10\,\mathrm{m}$, distance $26\,\mathrm{m}$
B2 (a) $3.03\,\mathrm{s}$, $-29.7\,\mathrm{m/s}$; (b) $20.4\,\mathrm{m}$, $2.04\,\mathrm{s}$, $4.08\,\mathrm{s}$
B3 $3.03\,\mathrm{s}$; $45.5\,\mathrm{m}$; $33.3\,\mathrm{m/s}$ at $63.2^\circ$ below the horizontal
B4 $1.02 \times 10^3\,\mathrm{m/s}$; $2.7 \times 10^{-3}\,\mathrm{m/s^2}$
B5 (a) $209\,\mathrm{km/h}$ at $16.7^\circ$ east of north; (b) $17.5^\circ$ west of north, ground speed $191\,\mathrm{km/h}$
B6 Train: straight up and down, flight time $1.63\,\mathrm{s}$. Ground: a parabola. The ball lands in the passenger's hand; ground distance $32.7\,\mathrm{m}$.

## Exam-style practice: answer key and brief solutions

**Key:** E1 D  E2 B  E3 B  E4 C  E5 A  E6 D  E7 D  E8 A  E9 E  E10 B

**E1 (D).** $v = \mathrm{d}x/\mathrm{d}t = 7.5t^2 - 8.0t + 3.0$ and $a = \mathrm{d}v/\mathrm{d}t = 15.0t - 8.0$, so $a(1.2) = 10.0\,\mathrm{m/s^2}$ (and $v(1.2) = 4.2\,\mathrm{m/s}$, which is the velocity, not the acceleration). Check by a second difference of $x(t)$.

**E2 (B).** $v^2 = v_0^2 + 2ad \Rightarrow a = \dfrac{20.0^2 - 8.0^2}{2(150)} = 1.12\,\mathrm{m/s^2}$. Check with the average speed $14.0\,\mathrm{m/s}$: $t = d/\bar{v} = 10.7\,\mathrm{s}$ and $a = \Delta v/t = 1.12$.

**E3 (B).** $t = (v - v_0)/a = 12.0/1.12 = 10.7\,\mathrm{s}$, using the previous result. Check: $t = 2d/(v_0 + v) = 300/28.0 = 10.7\,\mathrm{s}$.

**E4 (C).** Vertical motion only gives the time: $H = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{2H/g} = 3.03\,\mathrm{s}$. Horizontally $x = v_x t = (12.0)(3.03) = 36.4\,\mathrm{m}$.

**E5 (A).** Only the vertical component matters: $v_{y0} = 18.0 \sin 35^\circ = 10.3\,\mathrm{m/s}$, $h = v_{y0}^2/2g = 5.44\,\mathrm{m}$. Using the full speed gives $16.5\,\mathrm{m}$; using $\cos$ gives $11.1\,\mathrm{m}$.

**E6 (D).** I is false: the direction of $\vec{v}$ changes, so the velocity is not constant. II is true (centripetal acceleration). III is true: the circumference $2\pi r$ is covered at speed $v$. Answer: II and III only.

**E7 (D).** The crossing time depends only on the cross-stream speed: $t = 120/4.0 = 30\,\mathrm{s}$. In that time the current carries the boat $2.0 \times 30 = 60\,\mathrm{m}$ downstream. Check: the ground velocity $(2.0,\ 4.0)$ gives a slope $\Delta x/\Delta y = 0.50$, so $60\,\mathrm{m}$ in $120\,\mathrm{m}$.

**E8 (A).** Integrate each component: $v_x = 3.0t^2 = 12.0$ and $v_y = -2.0t = -4.0\,\mathrm{m/s}$ at $t = 2.0\,\mathrm{s}$. Speed $= \sqrt{12.0^2 + 4.0^2} = 12.6\,\mathrm{m/s}$. Substituting $t = 2.0$ into $\vec{a}$ without integrating is the common error.

**E9 (E).** $v = 6.0t - 12.0 = 0 \Rightarrow t = 2.0\,\mathrm{s}$. Then $x = 3.0(4.0) - 12.0(2.0) + 5.0 = -7.0\,\mathrm{m}$: the particle has turned back, and is on the negative side of the origin. The value $2.0$ is the time, not the position.

**E10 (B).** At the top $v = 0$: $t_{\text{up}} = v_0/g = 1.43\,\mathrm{s}$. The motion is symmetric (no air resistance), so the total time is $2v_0/g = 2.86\,\mathrm{s}$. Check: $y = v_0 t - \frac{1}{2}gt^2 = 0$ gives $t = 2v_0/g$.